

Statistical methods for computer science

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Descriptive Statistics

Where were we?

Statistical science is the science of developing and applying methods for collecting, analyzing and interpreting data.

Design phase:

- ✓ Definition of target population
- ✓ Choice of the sampling scheme
- ✓ Decision on the information to collect and how to measure them

Description phase:

Today Numerical description of data

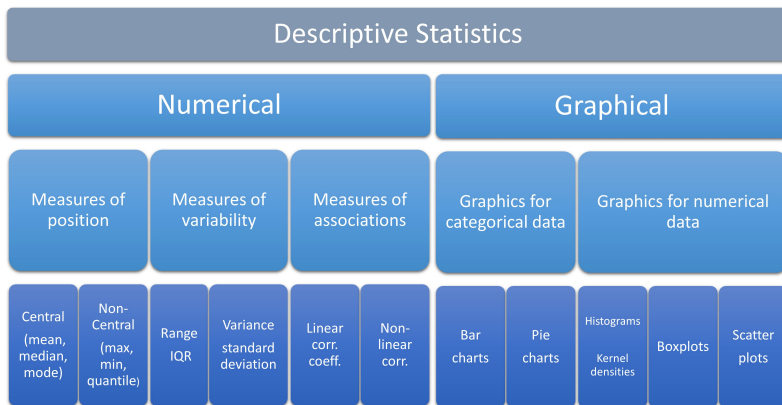
Today Graphical description of data

Course Outline

- Basic concepts
- **Descriptive statistics**
- Introduction to probability
- Introduction to random variables
- Discrete probability distributions
- Continuous probability distributions (+ R Lab)
- Sampling distributions
- Statistical inference: estimation (+ R Lab)
- Statistical inference: testing (+ R Lab)
- Linear model: estimation, inference, diagnostics (+ R Lab)

Introduction

Descriptive statistics summarize and describe the main features of a dataset, providing a way to understand and communicate the characteristics of the data



Running example

To illustrate the concepts covered in this lesson, we use the Income dataset, which lists annual income (in thousands of dollars) of 80 subjects classified by three categories of racial–ethnic status (black, hispanic, white). The dataset also contains information on the number of years of education of each subject.

	income	education	race
1	16	10	B
2	18	7	B
3	26	9	B
4	16	11	B
5	34	14	B
6	22	12	B
7	42	16	B
8	42	16	B
9	16	9	B
10	20	10	B

The dataset is based on a much larger sample taken by the U.S. Bureau of the Census and can be downloaded from the book website <https://stat4ds.rwth-aachen.de>

Descriptive Statistics

Numerical

Graphical

Measures of
position

Measures of
variability

Measures of
associations

Graphics for
categorical data

Graphics for numerical
data

Central
(mean,
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mode)

Non-
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(max,
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Measures of Central Tendency

They describe the center of the data. The mean, median, and mode are common measures of central tendency.

Given a sample of N observations of a variable Y from some population, the **sample mean** is

$$\bar{Y} = \frac{1}{N} \sum_{i=1}^N Y_i$$

Denoting with $Y_{(n)}$ the n -th observation of Y ordered in ascending order, the **median** is

$$\text{med } Y = Y_{(N+1)/2} \text{ (} N \text{ odd); } \text{med } Y = \frac{Y_{(N/2)} + Y_{((N/2)+1)}}{2} \text{ (} N \text{ even)}$$

The **mode** is the observation in $(Y_1, Y_2, \dots, Y_N)$ that appears more often (note that Y can have multiple modes or no mode).

Example

Given the integers 5, 10, 1, 3, 11, 20, 25, 5, 6, 5, 23, 2, 15, 2, 10, find:

- Sample mean:

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Given the integers 5, 10, 1, 3, 11, 20, 25, 5, 6, 5, 23, 2, 15, 2, 10, find:

- Sample mean:

$$\bar{Y} = \frac{5 + 10 + 1 + \cdots + 10}{15} = 9.53$$

- Median:

Example

Given the integers 5, 10, 1, 3, 11, 20, 25, 5, 6, 5, 23, 2, 15, 2, 10, find:

- Sample mean:

$$\bar{Y} = \frac{5 + 10 + 1 + \cdots + 10}{15} = 9.53$$

- Median: ordering the observations and taking the eight one from the beginning

$$1, 2, 2, 3, 5, 5, 5, \underbrace{6}_{\text{Median}}, 10, 10, 11, 15, 20, 23, 25$$

formally,

$$\text{med } Y = Y_8 = 6$$

- Mode:

Example

Given the integers 5, 10, 1, 3, 11, 20, 25, 5, 6, 5, 23, 2, 15, 2, 10, find:

- Sample mean:

$$\bar{Y} = \frac{5 + 10 + 1 + \cdots + 10}{15} = 9.53$$

- Median: ordering the observations and taking the eight one from the beginning

$$1, 2, 2, 3, 5, 5, 5, \underbrace{6}_{\text{Median}}, 10, 10, 11, 15, 20, 23, 25$$

formally,

$$\text{med } Y = Y_8 = 6$$

- Mode: the observation appearing more often (three times) is 5

Measures of Non-Central Tendency

*For Y sorted in ascending order, $Y_{(1)}$ is the **minimum** value of all observations and $Y_{(N)}$ is the **maximum** value.*

*The p -th **quantile** of N ordered values in ascending order is the value such that $p\%$ of the observations are smaller or equal to that point and $(1 - p)\%$ fall above it.*

- The 50% quantile is the median
- Special quantiles are the first and third **quartiles**, commonly denoted with Q_1 and Q_3 and defined respectively for $p = 0.25$ and $p = 0.75$
- The median and the quartiles split the data into four parts, each containing one-fourth of the observations

Measures of Non-Central Tendency

Denoting with $Q(p)$ the quantile of Y leaving $p\%$ of observations behind:

④ $Q(p) = Y_{(k)}$, where k is the nearest integer to pN .

* Remark: $\lceil pN \rceil$ denotes the ceiling of the product pN

Example

Given two samples of the same variable Y but with different lengths, $A = (2, 5, 6, 7, 8, 11, 12, 14, 23, 50, 56, 64)$ and $B = (2, 5, 6, 7, 8, 11, 12, 14, 23)$ find:

- First quartile:

Example

Given two samples of the same variable Y but with different lengths, $A = (2, 5, 6, 7, 8, 11, 12, 14, 23, 50, 56, 64)$ and $B = (2, 5, 6, 7, 8, 11, 12, 14, 23)$ find:

- First quartile: $pN_A = 0.25 \cdot 12 = 3$ and $pN_B = 0.25 \cdot 9 = 2.25$
 - ④ $Q_1^{(A)} = Y_{(3)} = 6$; $Q_1^{(B)} = Y_{(2)} = 5$
- Third quartile:

Example

Given two samples of the same variable Y but with different lengths, $A = (2, 5, 6, 7, 8, 11, 12, 14, 23, 50, 56, 64)$ and $B = (2, 5, 6, 7, 8, 11, 12, 14, 23)$ find:

- First quartile: $pN_A = 0.25 \cdot 12 = 3$ and $pN_B = 0.25 \cdot 9 = 2.25$
 - ④ $Q_1^{(A)} = Y_{(3)} = 6$; $Q_1^{(B)} = Y_{(2)} = 5$
- Third quartile: $pN_A = 0.75 \cdot 12 = 9$ and $pN_B = 0.75 \cdot 9 = 6.75$
 - ① $Q_3^{(A)} = Y_{(9)} = 23$; $Q_3^{(B)} = Y_{(7)} = 12$

R Example

```
> # Measures of central tendency
```

```
>
```

```
> mean(data$income)
```

```
[1] 37.525
```

```
> median(data$income)
```

```
[1] 30
```

```
> getmode(data$income)
```

```
[1] 30
```

```
>
```

```
> # Other measures of position
```

```
>
```

```
> qa <- quantile(data$income, probs = seq(0.25, 0.75, by = 0.25))
```

```
> qb <- quantile(data$income, probs = seq(0.05, 0.95, by = 0.05))
```

```
> data.frame(P = names(qa), Quartile = unname(qa))
```

```
> data.frame(P = names(qb), Quantile = unname(qb))
```

```
>
```

```
> # 6-number summary
```

```
>
```

```
> summary(data$income)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
16.00	22.00	30.00	37.52	46.50	120.00

	P	Quartile
1	25%	22.0
2	50%	30.0
3	75%	46.5

	P	Quantile
1	5%	17.9
2	10%	20.0
3	15%	20.0
4	20%	22.0
5	25%	22.0
6	30%	24.0
7	35%	26.0
8	40%	27.2
9	45%	30.0
10	50%	30.0
11	55%	32.0
12	60%	34.8
13	65%	40.0
14	70%	42.6
15	75%	46.5
16	80%	50.0
17	85%	56.0
18	90%	62.2
19	95%	82.1

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Measures of variability

They describe the spread or variability of the data around the center.

*The **range** is the difference between the maximum and minimum values of the observations in Y*

$$\text{Range} = Y_{(N)} - Y_{(1)}$$

- It provides a simple measure of the spread of the data
- Very sensitive to outliers

*The **interquartile range** is the difference between the third and the first quartiles of Y*

$$\text{IQR} = Q_3 - Q_1$$

- Less sensitive to outliers

Measures of variability

Given a sample of N observations of a variable Y from some population, the *sample variance* is

$$s^2 = \frac{\sum_{i=1}^N (Y_i - \bar{Y})^2}{N - 1}$$

and the *sample standard deviation* is

$$s = \sqrt{\frac{\sum_{i=1}^N (Y_i - \bar{Y})^2}{N - 1}}$$

- The sample variance is approximately an average of the squared deviations from the sample mean
- The sample standard deviation has the same unit of measurement of the original data (simpler to interpret)

Example

Given the integers 5, 10, 1, 3, 11, 20, 25, 5, 6, 5, 23, 2, 15, 2, 10, find:

- Range and IQR:

Example

Given the integers 5, 10, 1, 3, 11, 20, 25, 5, 6, 5, 23, 2, 15, 2, 10, find:

- Range and IQR: $Range = 25 - 1 = 24$, $Q_1 = Y_{(4)} = 3$,
 $Q_3 = Y_{(11)} = 11$, $IQR = Q_3 - Q_1 = 11 - 3 = 8$
- Sample variance:

Example

Given the integers 5, 10, 1, 3, 11, 20, 25, 5, 6, 5, 23, 2, 15, 2, 10, find:

- Range and IQR: $Range = 25 - 1 = 24$, $Q_1 = Y_{(4)} = 3$,
 $Q_3 = Y_{(11)} = 11$, $IQR = Q_3 - Q_1 = 11 - 3 = 8$
- Sample variance:

$$s^2 = \frac{(5 - 9.53)^2 + (10 - 9.53)^2 + \cdots + (10 - 9.53)^2}{14} = 61.84$$

- Sample standard deviation:

Example

Given the integers 5, 10, 1, 3, 11, 20, 25, 5, 6, 5, 23, 2, 15, 2, 10, find:

- Range and IQR: $Range = 25 - 1 = 24$, $Q_1 = Y_{(4)} = 3$,
 $Q_3 = Y_{(11)} = 11$, $IQR = Q_3 - Q_1 = 11 - 3 = 8$
- Sample variance:

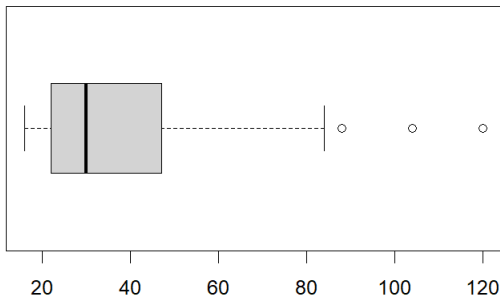
$$s^2 = \frac{(5 - 9.53)^2 + (10 - 9.53)^2 + \cdots + (10 - 9.53)^2}{14} = 61.84$$

- Sample standard deviation: $s = \sqrt{61.84} = 7.86$

R Example

```
> # Measures of variability
>
> c(Range = max(data$income)- min(data$income), IQR = qa[3]-qa[1])
  Range IQR.75%
  104.0    24.5
> var(data$income)
[1] 427.3665
> sd(data$income)
[1] 20.67284
> boxplot(data$income, horizontal = T, main = "Boxplot of annual income")
```

Boxplot of annual income



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Linear correlation coefficient

Given a sample of N observations for two variables X and Y , drawn from some population, their *sample covariance* is

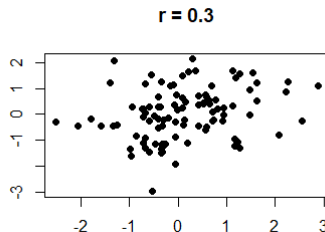
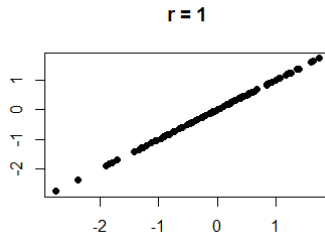
$$s_{XY} = \frac{\sum_{i=1}^N (Y_i - \bar{Y})(X_i - \bar{X})}{N - 1}$$

Given a sample for variables X and Y , their *sample linear correlation coefficient* is the ratio between their sample covariance and the product of their sample standard deviations,

$$r_{XY} = \frac{s_{XY}}{s_X s_Y}$$

Linear correlation coefficient

- it is also named “Pearson correlation coefficient” after Karl Pearson
- it is a normalized measure of the covariance as it takes value in $[-1, 1]$ (1 means perfect positive linear relation, -1 means perfect negative linear relation)
- if the variables X and Y are independent, $r_{XY} = 0$ but the reverse is not true (a zero linear correlation does not imply that the variables are independent)

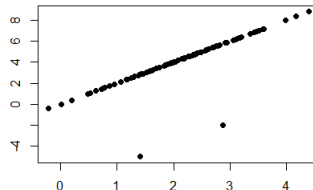


Linear correlation coefficient

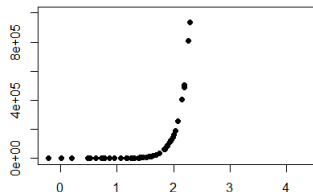
Pitfalls:

- it is very sensitive to outliers
- being a measure of *linear* dependence, when $r_{XY} = 0$ it means that X and Y are *linearly* unrelated but the two variables may still have a non-linear relation

$$y = 2(x), r = 0.85$$



$$y = \exp(6x), r = 0.33$$



Categorical data

As the name says, all the numerical measures of position and variability seen so far are defined for numerical data. What about categorical data?

- The **mode** is the only measure of central tendency that can be used for categorical data
- The median, mean, variance and range are not meaningful for categorical data: **counts** and **percentages** can be used instead to describe, respectively, the number and the proportion of observations falling in each category

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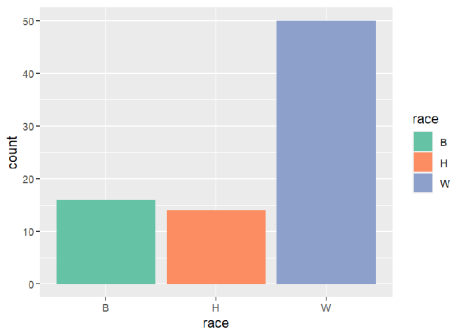
Each category is represented by a bar, and the height of each bar represents the counts or the proportion of that category in the data

```
> # Finding unique values for 'race' (categories)
> unique(data$race)
[1] "B" "H" "W"
> getmode(data$race)
[1] "W"
>
> # Counts (how many observation in each category?)
> table(data$race)

  B  H  W
16 14 50

>
> # Proportions (% of observations in each category)
> table(data$race)/nrow(data)

      B      H      W
0.200 0.175 0.625
```

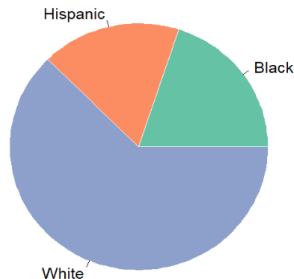


Pie charts

Each category is represented by a slice of the pie, and the size of each slice corresponds to the proportion of that category in the data

From R help page:

- Pie charts are a bad way of displaying information (the eye is good at judging linear measures and bad at judging relative areas)
- Data that can be shown by pie charts always can be shown by a bar chart



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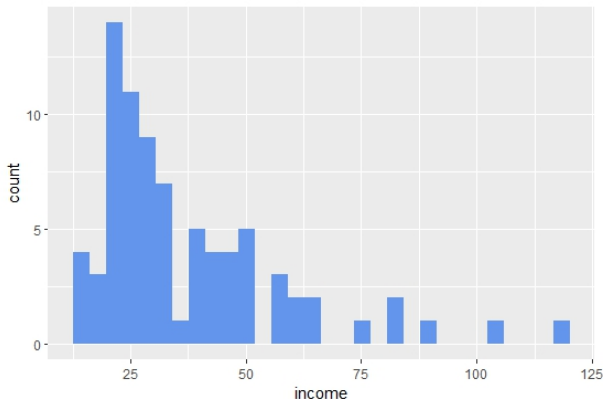
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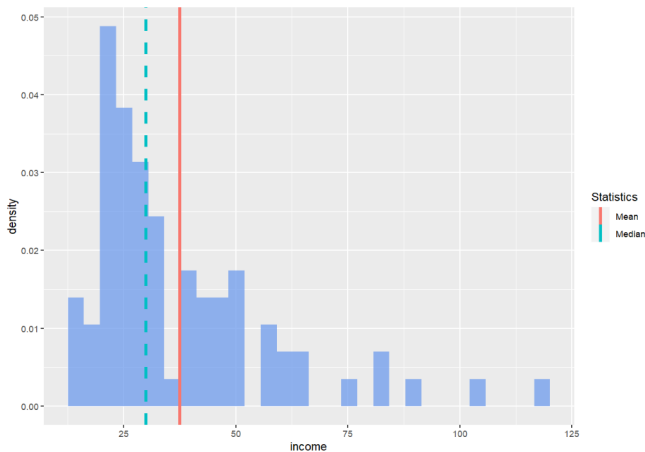
Histograms

The range of values is divided into intervals or “bins” usually of equal size and with height given by the counts (or the proportion of observations) in each bin.



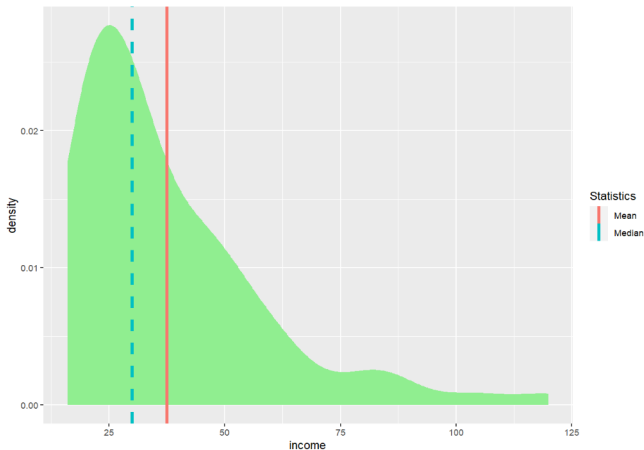
Histograms

Starting from proportions, when the height of each bin is also divided by interval width, the total area of the histogram is 1, in which case it can be considered an approximate density of the data



Density

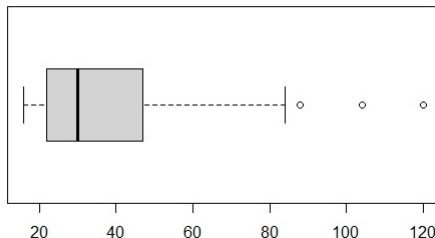
If we imagine to increase the number of intervals indefinitely, the shape of the histogram would gradually approach a smooth curve leading to an estimation of the probability density function



Box plots

They provide a representation of the 5-number summary: the box represents the IQR, the line inside the box is the median, whiskers extend from the edges of the box to 1.5 times the IQR (or to the min/max values if they fall within the range)

Boxplot of annual income

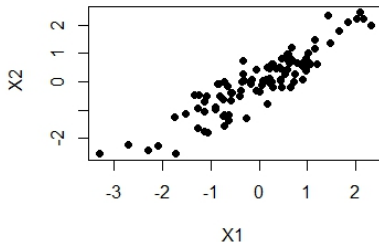


Example: recall that $IQR = 24.5$, $Q_1 = 22$, $Q_3 = 46.5$, $Min = 16$, $Max = 120$. Upper whisker ends at $Min\{Max(Y); Q_3 + 1.5 \cdot IQR\} = 83.25$, lower whisker ends at $Max\{Min(Y); Q_1 - 1.5 \cdot IQR\} = 16$

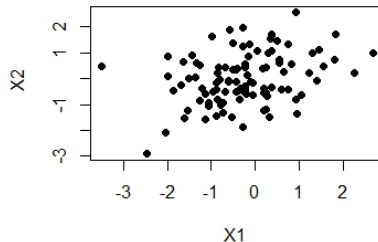
Scatter plots

For a pair of variables, we can plot values for one variable on the horizontal axis and for the other variable on the vertical axis, obtaining a scatter plot (useful to portray the correlation between the two variables)

Correlation = 0.9



Correlation = 0.4



Where are we?

Statistical science is the science of developing and applying methods for collecting, analyzing and interpreting data.

Design phase:

- ✓ Definition of target population
- ✓ Choice of the sampling scheme
- ✓ Decision on the information to collect and how to measure them

Description phase:

- ✓ Numerical description of data
- ✓ Graphical description of data

Inference: next lecture!